MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION,

MUMBAI

A Project Report on

Gas Agency Stock Management System

Submitted in partial fulfillment for the award of

Diploma in Computer Engineering

By

Deore Kishori Nitin

Desale Bhimika Sudhir

Desale Mayuri Suresh

Patil Neha Nitin

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

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CERTIFICATE

This is to certify that,

Sr. No.	Roll No.	Name	Enrollment Number	Exam Seat number
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3	28	Desale Mayuri Suresh	23610960202	405591
4	68	Patil Neha Nitin	23610960251	405631

from **S.S.V.P.S's B. S .Deore Polytechnic, Dhule** have completed Capstone Project of final year having title **"Gas Agency Stock Management System"** during the academic year 2025-2026. The project is completed in a group consisting of four persons under the guidance of the Faculty Guide.

Date : **Place:** Dhule

Signature: **Signature:**

Name : Mr. N. S. Sharma **Name :** Mr. Nishad D. Patel

Guide **H.O.D.**

Signature: **Signature:**

Name : **Name :** Prof. S. A. Hire

External **Principal**

Acknowledgment

We want to thanks to the Owner and gas agency employees who challenges inspired us to develop this application.

We want to thanks our project guide Mr. N. S. Sharma, for his constant support and important contributions. His support and confidence in us kept us motivated throughout journey and help us do our best.

We also want thank our capstone project co-ordinator, Mrs. R. S. Patil mam for this able advice and help. He played a big part in shaping our project and his hard work greatly us success.

We're grateful to our Head of the Department, Mr. N. D. Patel, for always being there for management of S. S. V. P. S. B. S. Deore Polytechnic.

Finally we would like to express our deepest gratitude to our families and friends for beginning there for us when we needed them the most.

Deore Kishori Nitin

Desale Bhumika Sudhir

Desale Mayuri Suresh

Patil Neha Nitin

Abstract

The proposed system is **"**Gas Agency Stock Management System**."** It is a web application designed to manage the stock and daily operations of a gas agency. In many gas agencies, stock details, cylinder records, and customer information are often managed manually, which can lead to errors and difficulty in tracking data. This system works as a digital solution to manage all gas agency records in an organized way. The main goal of this system is to simplify the management of cylinder stock, suppliers, and deliveries. It helps agency staff to easily add, update, and monitor the availability of different types of gas cylinders such as ssssss, and report generation, which help the agency staff to monitor operations efficiently. The system also keeps proper records of cylinder transactions, making it easier to retrieve information whenever needed. The Gas Agency Stock Management System provides a user-friendly interface so that staff and administrators can easily manage and monitor the stock details. Overall, the system helps improve efficiency, saves time, and ensures better management of gas cylinder inventory in the agency.

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Chapter 1
Introduction and Background of Industry

1.1 Introduction:

The Liquefied Petroleum Gas (LPG) distribution industry plays a vital role in the daily lives of households, commercial establishments, and small industries. In India, LPG is one of the most widely used and essential fuels for cooking purposes due to its efficiency, affordability, cleanliness, and ease of use. Gas agencies act as the bridge between oil marketing companies and end consumers by ensuring timely booking, storage, distribution, and delivery of LPG cylinders.

A gas agency is responsible for managing customer registrations, issuing new gas connections, handling refill bookings, maintaining stock records, organizing cylinder delivery, collecting payments, and resolving customer complaints. With the growing number of customers and increasing demand for faster services, traditional manual systems are no longer sufficient. Therefore, digital management systems are becoming necessary to improve accuracy, reduce workload, and enhance customer satisfaction.

The LPG gas distribution industry has become more organized and technology-driven over time. Today, agencies use computerized records, customer databases, delivery tracking, and stock management systems to ensure smooth business operations. A Gas Agency Management System is highly useful in this industry because it helps in maintaining proper records of filled and empty cylinders, customer details, booking status, delivery staff work, and daily sales reports.

1.2 Background of the Industry:

The Liquefied Petroleum Gas (LPG) distribution industry is an essential service sector in India, providing clean, efficient and convenient fuel for domestic, commercial and small industrial use. LPG is widely used in households, hotels, restaurants and small businesses due to its ease of use, affordability and environmental benefits.

In India, LPG distribution is mainly managed through authorized gas agencies such as Bharat Gas, Indane Gas and HP Gas. These agencies handle important activities like customer registration, refill booking, stock management, cylinder delivery, billing and customer support.

Traditionally, many gas agencies managed these operations manually, which often caused errors, delays and difficulties in record maintenance.

With the growth in customer demand and advancement in technology, computerized systems have become necessary for better management. A Gas Agency Management System helps in maintaining customer records, tracking bookings, managing stock, handling billing and generating reports efficiently. Therefore, the LPG gas agency industry provides a practical and suitable domain for developing this project.

Capstone Project

A capstone project is the final project that students complete at the end of their course. It helps students apply the knowledge and skills they have learned during their studies to solve real-world problems.

Our capstone project is Gas Agency Management System. The main purpose of this system is to make the work of a gas agency easier and more organized. In many gas agencies, stock records, billing, staff details, and delivery information are managed manually, which can lead to errors and takes more time.

This system provides a digital solution to manage gas agency operations. It includes different modules such as Stock Management, Billing Generation, Staff Management, and Delivery Management. Through this system, the agency can easily track cylinder stock, generate bills, manage staff records, and monitor deliveries.

The project helps improve accuracy, efficiency, and record management in gas agencies. It also gives students practical experience in developing a real-world application using modern technologies.

Therefore, this capstone project helps us integrate our learning and prepares us for our future careers by giving us experience in solving real-life problems using technology.

Chapter 2

Literature Survey and Specification

2.1 Study of existing system / Review of research paper:

To build an effective Gas Agency Management System, it is crucial to first understand the current working system of gas agencies and to review the existing methods used for managing LPG distribution in local agencies. The study of the existing system helps to identify the practical difficulties faced by gas agency staff, delivery staff, customers and owner/admin during daily operations. It also helps to understand how a computerized system can improve the efficiency, transparency and reliability of the gas agency process.

Our examination of existing gas agency management practices has revealed several common challenges. Many gas agencies still rely on manual and paper-based processes for customer registration, gas refill booking, stock entry, billing and delivery management. This traditional approach often leads to delays, errors, confusion and lack of transparency. Managing customer details, refill booking records, filled and empty cylinder stock, delivery schedules and payment records manually becomes a difficult and time-consuming task in the absence of a dedicated system. Moreover, gas agency-related data is often scattered across different registers, files and notebooks, making it difficult to maintain proper records and generate reports whenever required.

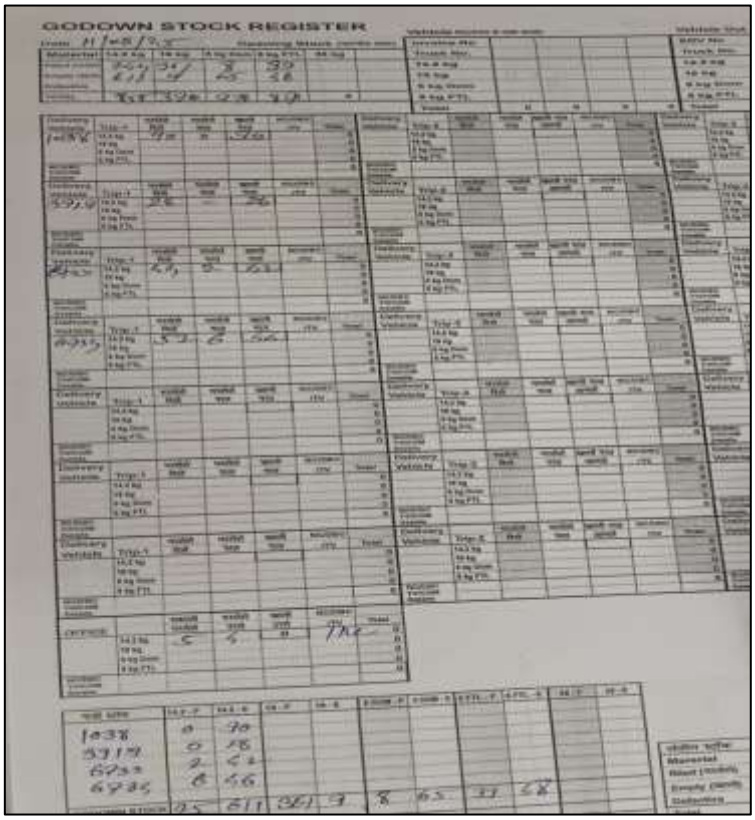
We have taken a closer look at how gas agencies currently manage their operations. Here's what we found:

Admin / Owner:

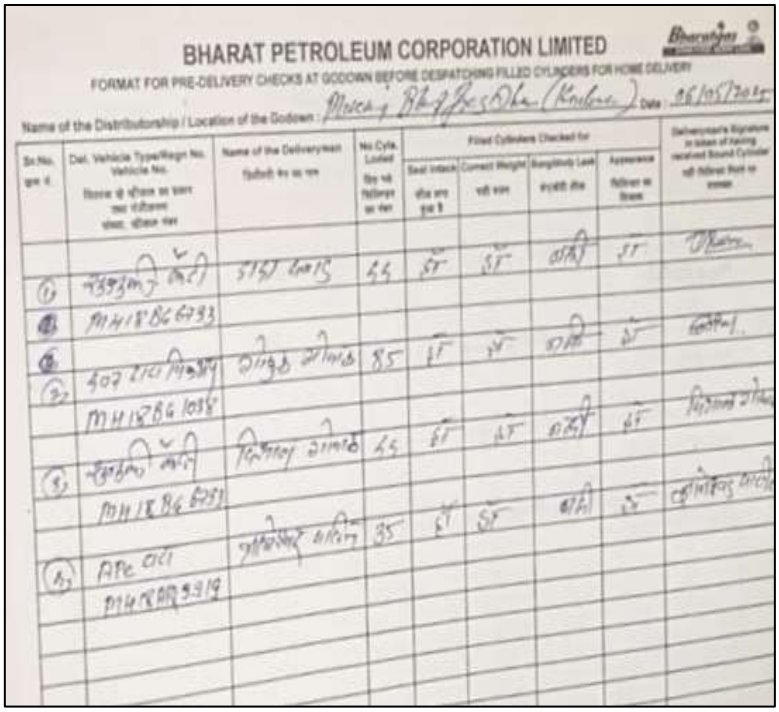
Manage complete gas agency operations

- Admin / Owner currently manages customer records manually using registers or files.
- Admin / Owner currently handles refill bookings manually by phone calls or written booking entries.
- Admin / Owner currently maintains cylinder stock details such as filled cylinders, empty cylinders and damaged cylinders in separate registers.
- Admin / Owner currently manages delivery staff records and assigns delivery work manually.
- Admin / Owner currently checks booking status, delivery status and payment records manually.
- Admin / Owner currently generates daily or monthly reports manually, which is time-consuming and may lead to mistakes.

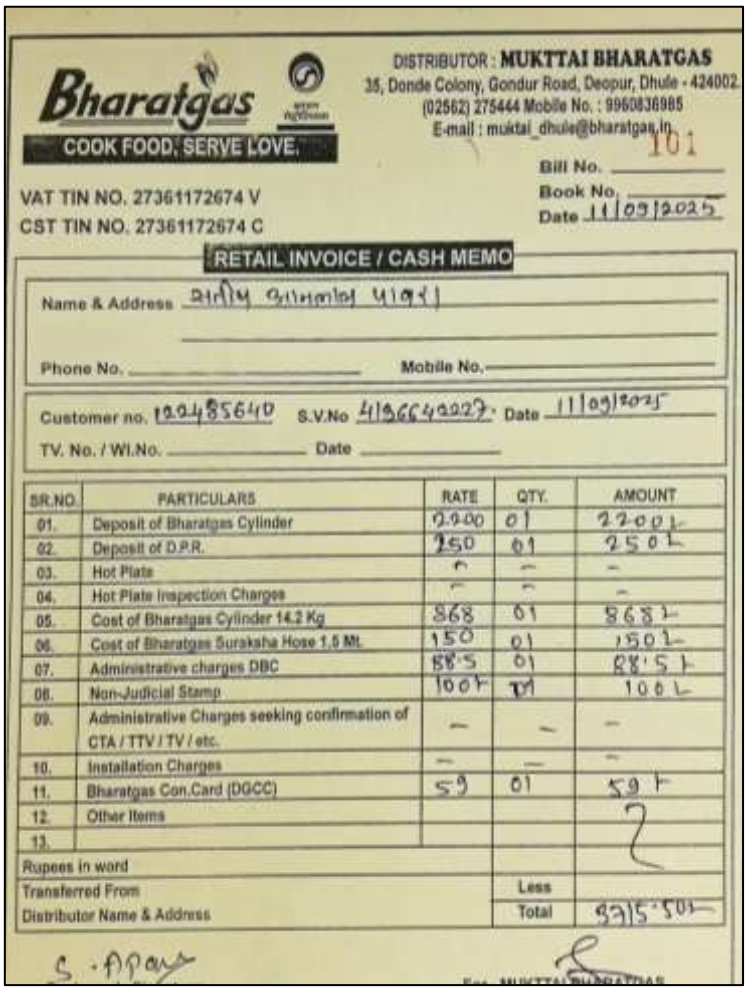
- Admin / Owner currently faces difficulty in tracking overall agency performance and stock availability accurately.



Godown Stock Register is used to record daily LPG cylinder stock details in the agency. It maintains opening stock, delivery vehicle-wise dispatch, office stock, and closing stock. It helps in proper stock management and tracking of cylinder movement.



Pre-Delivery Check Register is used to verify LPG cylinders before dispatch for delivery. It records vehicle details, deliveryman details, loaded cylinders, and safety checking information. It ensures safe and proper delivery of cylinders to customers.



Bharatgas

COOK FOOD, SERVE LOVE.

DISTRIBUTOR : MUKTTAI BHARATGAS
35, Donde Colony, Gondur Road, Deopur, Dhule - 424002.
(02562) 275444 Mobile No. : 9960836985
E-mail : muktai_dhule@bharatgas.in

Bill No. 101
Book No.
Date 11/09/2025

VAT TIN NO. 27361172674 V
CST TIN NO. 27361172674 C

RETAIL INVOICE / CASH MEMO

Name & Address [illegible]

Phone No.
Mobile No.

Customer no. 120485640 S.V.No 4136642027 Date 11/09/2025
TV. No. / WI.No. Date

SR.NO.	PARTICULARS	RATE	QTY.	AMOUNT
01.	Deposit of Bharatgas Cylinder	2200	01	2200/-
02.	Deposit of D.P.R.	250	01	250/-
03.	Hot Plate	-	-	-
04.	Hot Plate Inspection Charges	-	-	-
05.	Cost of Bharatgas Cylinder 14.2 Kg	868	01	868/-
06.	Cost of Bharatgas Suraksha Hose 1.5 Mt.	150	01	150/-
07.	Administrative charges DBC	88.5	01	88.5/-
08.	Non-Judicial Stamp	100/-	01	100/-
09.	Administrative Charges seeking confirmation of CTA / TTV / TV / etc.	-	-	-
10.	Installation Charges	-	-	-
11.	Bharatgas Con.Card (DGCC)	59	01	59/-
12.	Other Items			
13.				
Rupees in word				
Transferred From			Less	
Distributor Name & Address			Total	3715.50/-

S. Apar

For MUKTTAI BHARATGAS

This document is a Retail Invoice / Cash Memo issued by Bharatgas for providing a new LPG gas connection to the customer. The invoice contains details such as distributor information, customer name, customer number, mobile number, and billing date. It includes various charges like cylinder security deposit, D.P.R deposit, cost of a 14.2 kg LPG cylinder, Suraksha hose pipe cost, administrative charges, non-judicial stamp charges, and connection card charges. All these components together form the total payable amount of ₹3715.50. This invoice serves as an official payment receipt and proof of registration for the LPG gas connection provided to the customer.

Study of Existing Gas Agency Management Applications

Table 1 Study of Existing Gas Agency Management Applications

Sr No.	Application	Description
1.	Indane Gas Online Services	This system provides services such as customer registration, refill booking, booking status check, subsidy information, complaint registration, and online payment.
2.	Bharat Gas Online Portal	This system provides functionalities like new connection request, refill booking, online payment, delivery status, complaint support, and customer profile management.

2.2 Limitations of existing system / Problems discussed in research papers :

i. Reliance on manual and paper-based processes

ii. Time-consuming customer record management

iii. Errors in refill booking and billing

iv. Improper stock monitoring

v. Lack of real-time delivery tracking

2.3 Problem Identification / Need of a system :

i. Manual management of customer records, gas connections, and refill bookings is difficult and time-consuming.

ii. Proper tracking of filled, empty, and damaged LPG cylinder stock is challenging in the existing system.

iii. Manual handling of cylinder delivery schedules and delivery status leads to delays and inefficiency.

iv. Monitoring bookings, payments, stock, and complaints is difficult due to scattered paper-based records.

v. Hence, a computerized Gas Agency Stock Management System is required to improve accuracy, efficiency, and service quality.

2.4 Problem definition:

Gas agencies face difficulties in managing cylinder stock, customer records, delivery tracking, and billing using manual systems. These traditional methods often lead to errors, data mismanagement, delays, and inefficient operations. There is a need for an automated

Gas Agency Stock Management System to improve accuracy, reduce manual work, and ensure efficient and reliable management of agency operations.

2.5 Specification:

- **Hardware Specification:**

  Operating System: Windows 10 (64 bit) 4.00GB

  Processer: Intel(R) Core(TM) i3-3220 CPU @3.30 GHz 3.30 GHz Ram: 4.00GB

- **Software specification:**

  Server: XAMPP 8.0.28

  Database: MySQL 5.2.1

Chapter 3

Scope of Project

The scope of this project includes developing a Gas Agency Management System that helps manage the daily operations of a gas agency digitally. The system is designed to make stock management, billing, staff management, and delivery tracking easier and more efficient.

The system will support different user roles such as Admin, Staff, and Delivery Person. The Admin will have full control of the system, including managing stock, staff records, and monitoring all operations. Staff members will be responsible for billing and updating customer or cylinder records. Delivery persons will update delivery status and manage cylinder distribution.

The system will allow the agency to maintain accurate records of cylinders, generate bills quickly, track deliveries, and monitor stock levels. This will reduce manual work, minimize errors, and improve overall efficiency of the gas agency.

Non-functional requirements of the system include secure login authentication, easy navigation, real-time updates of stock and delivery status, role-based access control, and a user-friendly interface.

The main goal of this project is to provide a digital solution for gas agency management, improve record keeping, and ensure smooth coordination between admin, staff, and delivery personnel.

User Roles: *Admin (Agency Manager)*, *Staff (Office Staff)*, *Delivery Person*

- Cylinder stock management
- Billing generation
- Staff management
- Delivery tracking
- Managing cylinder records
- Updating stock levels
- Generating customer bills
- Managing staff details
- Tracking delivery status
- Viewing agency records
- Real-time stock updates
- Secure login authentication
- Role-based access control
- User-friendly interface

Chapter 4
Methodology

4.1 User Requirement:

Functional Requirement:

There are three user roles for the system: Admin (Agency Manager), Staff, and Delivery Person.

i. Requirement of Admin (Agency Manager):

- Admin able to view total cylinder stock and its details.
- Admin can add, update, and delete cylinder records.
- Admin able to manage staff details (add, update, remove staff).
- Admin can monitor all billing records generated by staff.
- Admin shall be able to track delivery status of cylinders.
- Admin able to view reports such as stock report, sales report, and delivery report.
- Admin shall be able to manage customer records.
- Admin able to control overall system activities.

ii. Requirements of Staff (Office Staff):

- Staff shall be able to login with unique username and password.
- Staff shall be able to generate bills for customers.
- Staff shall be able to view and update cylinder stock details.
- Staff shall be able to manage customer information.
- Staff shall be able to view previous billing records.
- Staff shall be able to update order details.
- Staff shall be able to coordinate with delivery person for orders.

iii. Requirements of Delivery Person:

- Delivery person shall be able to login with username and password.
- Delivery person shall be able to view assigned delivery orders.
- Delivery person shall be able to update delivery status (delivered / pending).
- Delivery person shall be able to track delivery details.
- Delivery person shall be able to view customer address and contact details.
- Delivery person shall be able to update delivery records in system.

Non Functional Requirement:

- System should implement secure user authentication and authorization to protect sensitive data like billing, customer details, and staff records.
- System should provide real-time updates of cylinder stock, billing records, and delivery status.
- System should support role-based access control so that Admin, Staff, and Delivery Person get permissions according to their roles.
- System should have a user-friendly and self-explanatory interface so that users can easily understand and operate it.
- System should be easy to use even for users with less technical knowledge.
- System should ensure data accuracy and consistency in stock, billing, and delivery records.
- System should provide fast performance and quick response time during operations like billing and stock updates.
- System should be reliable and available whenever required without frequent failures.
- System should allow easy maintenance and future updates.

4.2 Block Diagram of System:

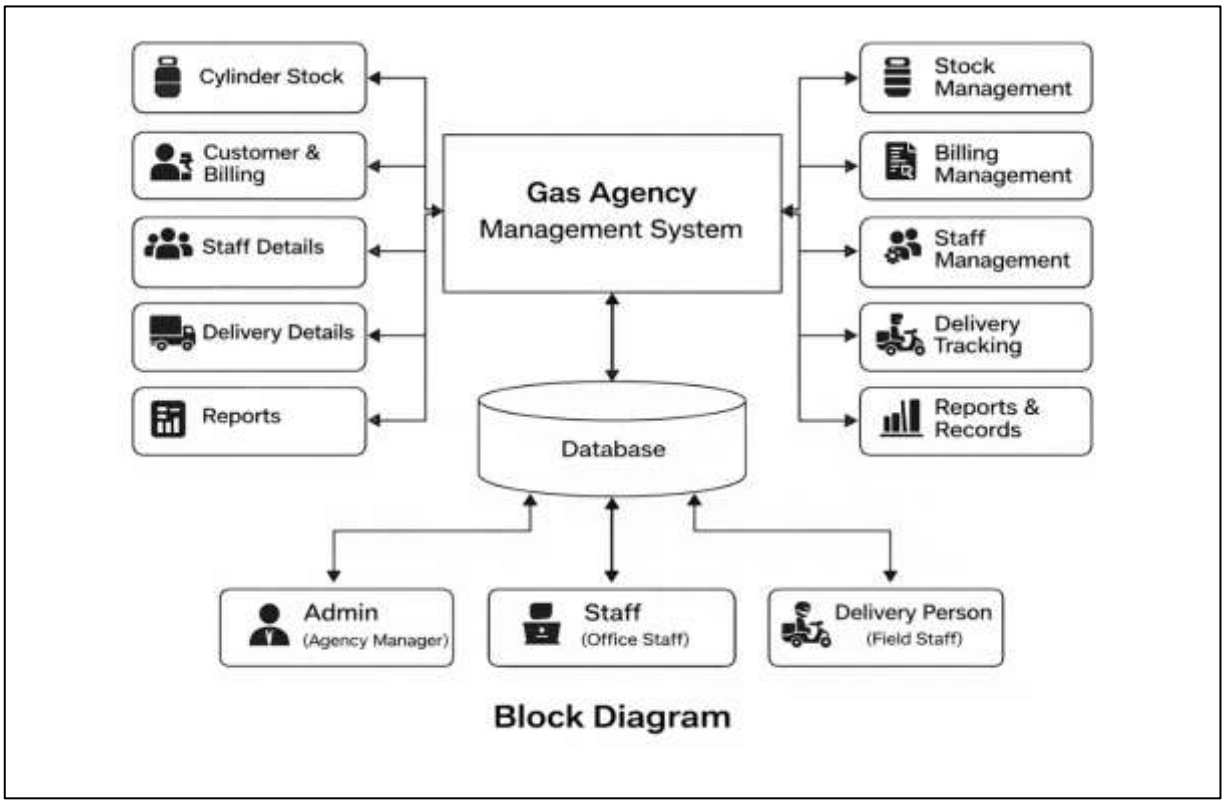


Figure 4.1 Block diagram of Gas Agency Management system

Figure 4.1 shows the proposed system diagram of Gas Agency Management System, where the diagram is divided into three parts: Input, Output, and Database, which helps in managing overall gas agency operations efficiently.

Input: Input includes all the data required for the system to function properly. Cylinder Stock Details consist of cylinder ID, type (domestic/commercial), quantity, supplier name, and stock status. Customer Details include customer name, address, mobile number, and connection details. Billing Details contain bill number, customer name, cylinder type, quantity, price, and date of billing. Staff Details include staff name, role, contact number, and login credentials. Delivery Details consist of delivery ID, customer address, delivery date, and delivery status (delivered/pending).

Output : System generates billing receipts for customers. System provides stock reports showing available and low stock cylinders. System generates delivery status reports (delivered / pending orders). System provides staff details and performance records. System allows users to search and filter data for easy access. System provides visual representation (charts/reports) to monitor stock and sales easily.

4.3 Action Plan:

Steps followed to develop Gas Agency Management System:

- Finalization of project team and understanding project requirements
- Identification of project domain (Gas Agency Management)
- Study of existing manual system used in gas agencies
- Collection of requirements such as stock management, billing, staff, and delivery
- Finalization of project idea with guidance from project guide
- Literature survey and analysis of similar systems
- Project identification and defining system specifications
- Preparation of detailed methodology to solve the problem
- Preparation of project planning report
- Designing system architecture and workflow
- Designing user interface (UI) for Admin, Staff, and Delivery Person
- Designing database structure and creating database
- Development of modules (Stock, Billing, Staff, Delivery)
- Programming and coding of system
- Database connectivity implementation
- Testing of system for errors and performance
- Installation and execution of the project
- Modifications based on suggestions from guide

- Preparing project presentation (PPT)
- Preparing draft of complete project report
- Making corrections in project report and presentation
- Final submission of project report and term work

4.4. Technology use to develop project:

Structure: React

Design: CSS

Validation: JAVASCRIPT

Database: MongoDB

Chapter 5
Details of designs, working and processes

5.1 System Flow Diagram

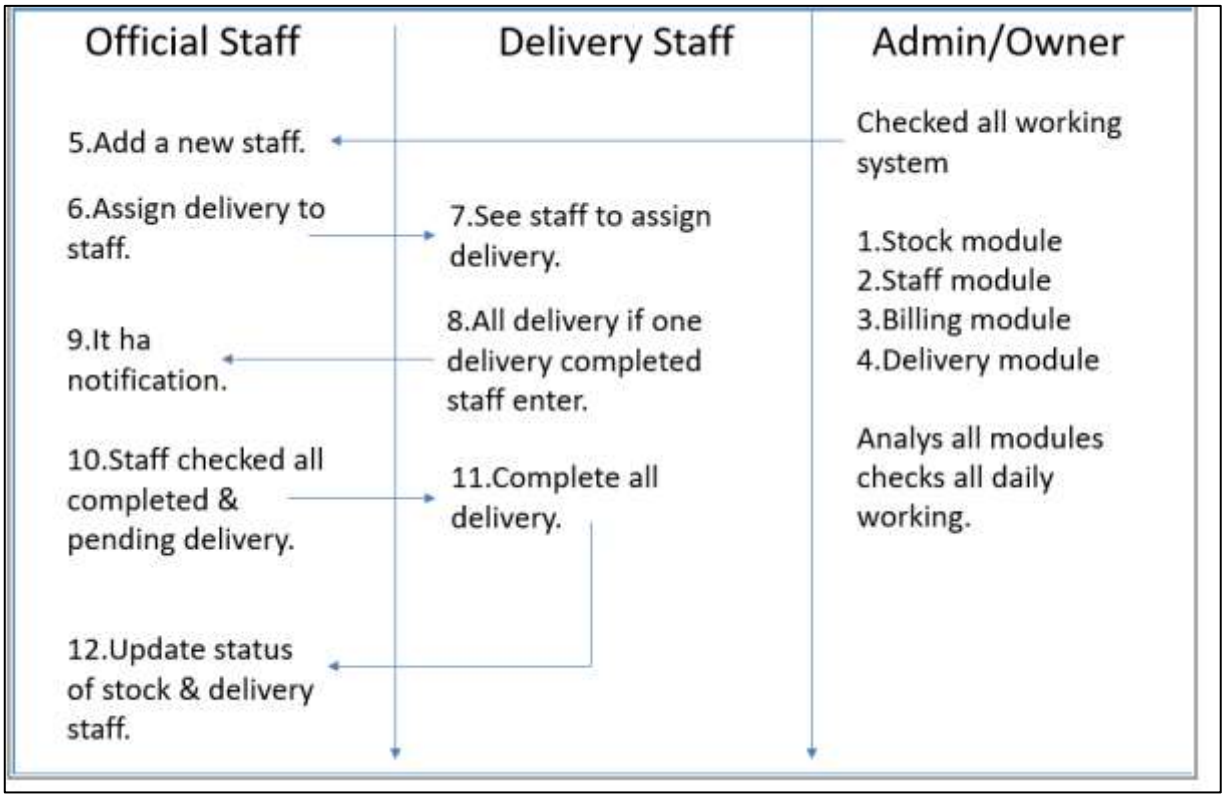


Figure 5.1 Flow diagram of Gas Agency Stock Management system

Figure 5.1 shows the flow diagram of the Gas Agency Stock Management System in which three main user roles are displayed: Admin/Owner, Official Staff, and Delivery Staff. The diagram explains the step-wise responsibilities and interaction of each user within the system. Initially, the Admin/Owner manages the overall system setup and monitors daily operations. The admin maintains all important modules such as Stock Management, Staff Management, Billing Management, and Delivery Management. Admin verifies system activities and ensures smooth workflow execution. After system initialization, Official Staff performs operational activities. Official staff registers new staff members and assigns delivery tasks to delivery staff based on customer orders and stock availability. The staff continuously monitors delivery status including completed and pending deliveries. Notifications regarding delivery updates are received automatically in the system. Official staff also updates stock information and delivery records after task completion. Delivery Staff plays an important execution role in the system. Delivery staff logs into the system, views assigned deliveries, and performs cylinder delivery to customers. After completing each delivery, delivery staff updates delivery status in the system. This process is repeated daily for all delivery tasks assigned by the office. Throughout the working cycle, the system automatically updates stock quantity, billing records, and delivery information. Notifications are generated whenever delivery or billing activities are completed. Admin/Owner continuously analyzes system performance and checks daily working reports.

5.2 Site Map:

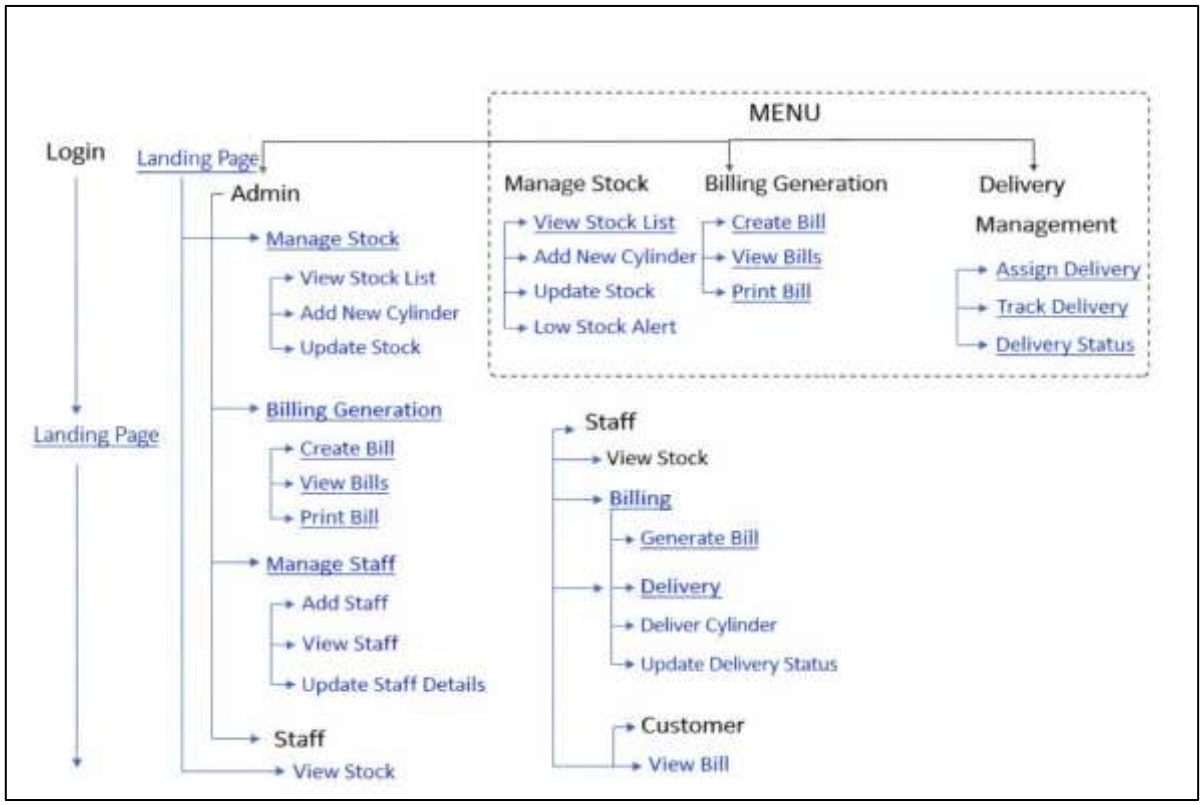


Figure 5.2 Site Map of Gas Agency StockManagement system

Figure 5.2 shows the site map of project work monitoring system from left to right.1st page displayed to all user is landing page and this web page separate every user according its role as Student, mentor and admin. According user role login page is display and authorized user is allowed to access their dashboard. At admin dashboard menu design to redirect admin for manage student, manage mentor, manage timeline, manage group. Similarly at mentor dashboard and student dashboard menu is designed to redirect for manage timeline defined by admin.

5.3 Admin Module Flow:

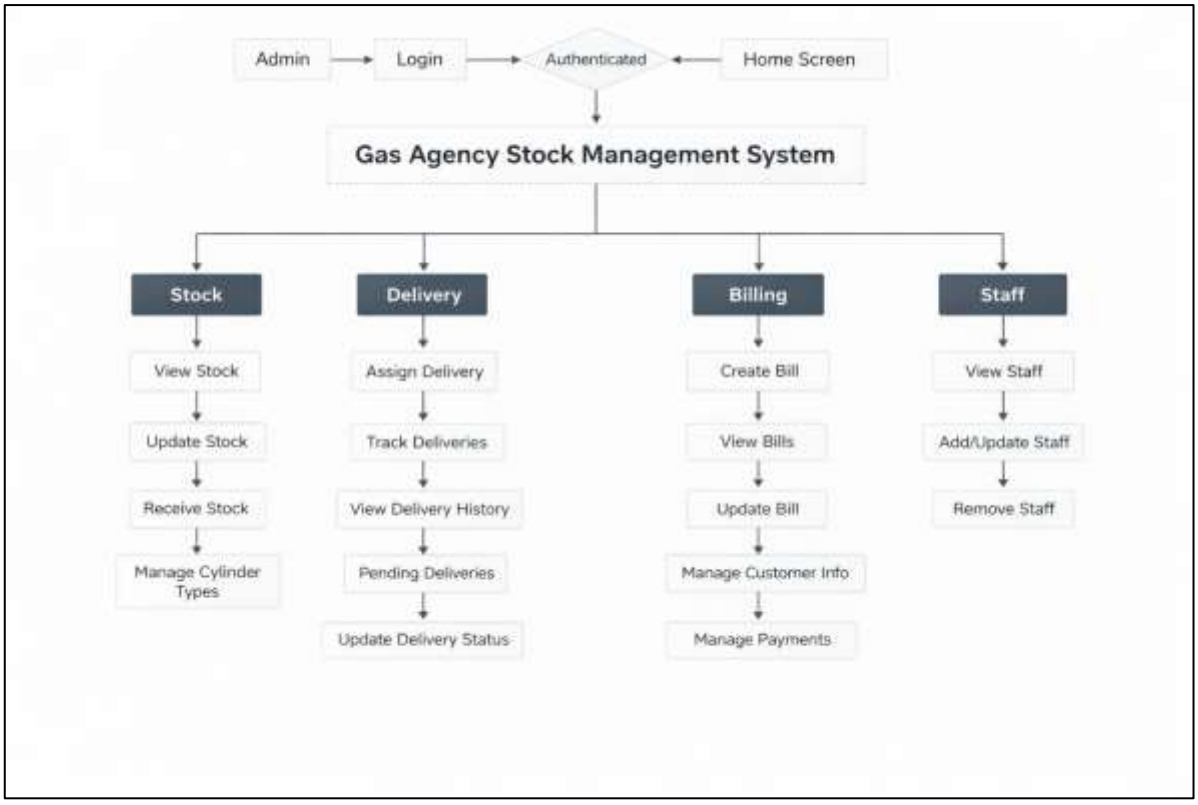


Figure 5.3 Admin module flow

Figure 5.3 shows the Admin module flow of the Gas Agency Stock Management System. From the main menu, the admin is redirected to manage key functionalities such as Stock Management, Delivery Management, Billing Management, and Staff Management. In the Stock Management module, the admin can view current stock details, update stock quantities, record received stock, and manage different types of gas cylinders. This helps maintain accurate inventory levels and ensures smooth operations. In the Delivery Management module, the admin can assign deliveries to staff, track delivery status, view delivery history, and manage pending deliveries. This module ensures timely and efficient delivery of gas cylinders to customers. In the Billing Management module, the admin is able to create new bills, view existing bills, update billing information, manage customer details, and handle payment records. This ensures proper financial tracking and transparency. In the Staff Management module, the admin can view staff details, add or update staff information, and remove staff records when required. This helps in maintaining proper workforce management. Overall, the system enables the admin to efficiently monitor and control all major operations including stock, delivery, billing, and staff activities within the gas agency.

5.4 Official Staff Module Flow:

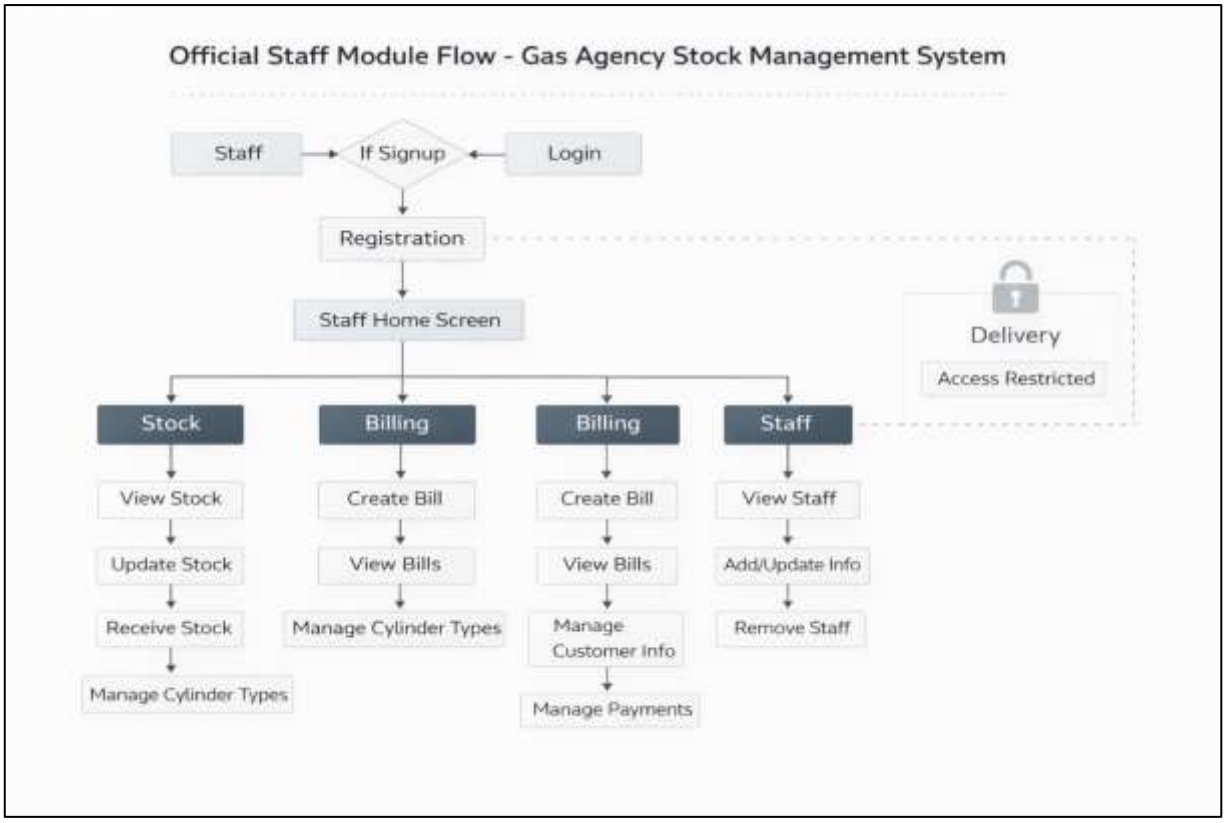


Figure 5.4 Official Staff module flow

Figure 5.4 shows the Official Staff module flow of the Gas Agency Stock Management System. The staff user can register or log in to the system and, after successful authentication, is redirected to the staff home screen. From there, the staff can access Stock, Billing, and Staff management modules. In the Stock module, staff can view and update stock details, while in the Billing module they can create and manage bills and customer payment information. The Staff module allows viewing and updating staff details. The Delivery module is restricted and not accessible to staff, ensuring role-based access control and system security.

5.5 Delivery Staff Module Flow:

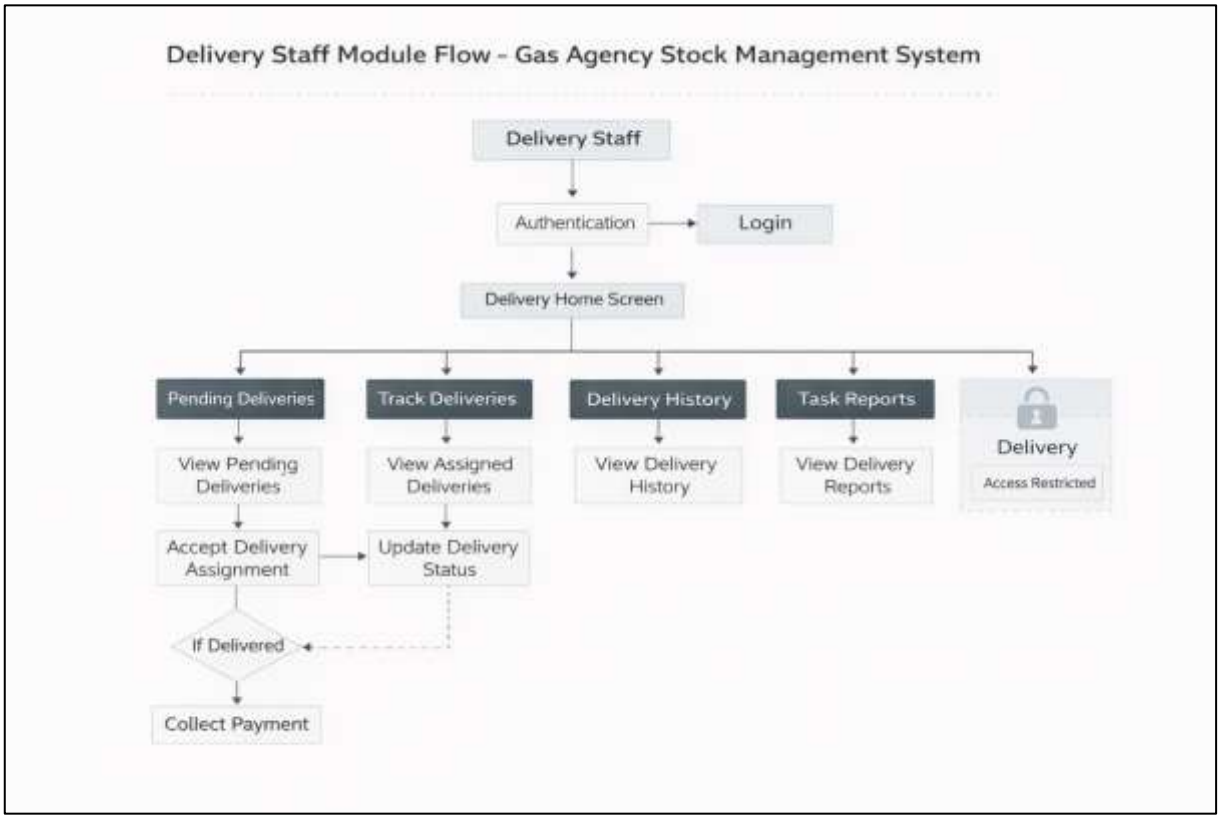


Figure 5.5 Delivery Staff module flow

Figure 5.5 shows the Delivery Staff module flow of the Gas Agency Stock Management System. The delivery staff can register or log in to the system and, after successful authentication, is redirected to the delivery home screen. From there, the staff can view pending deliveries, accept assigned tasks, and update delivery status through the tracking module. After completing a delivery, the staff can collect payment and record the transaction. The system also allows viewing delivery history and task reports for performance tracking, while restricting access to non-delivery modules to ensure secure and role-based operations.

5.6 Er Diagram:

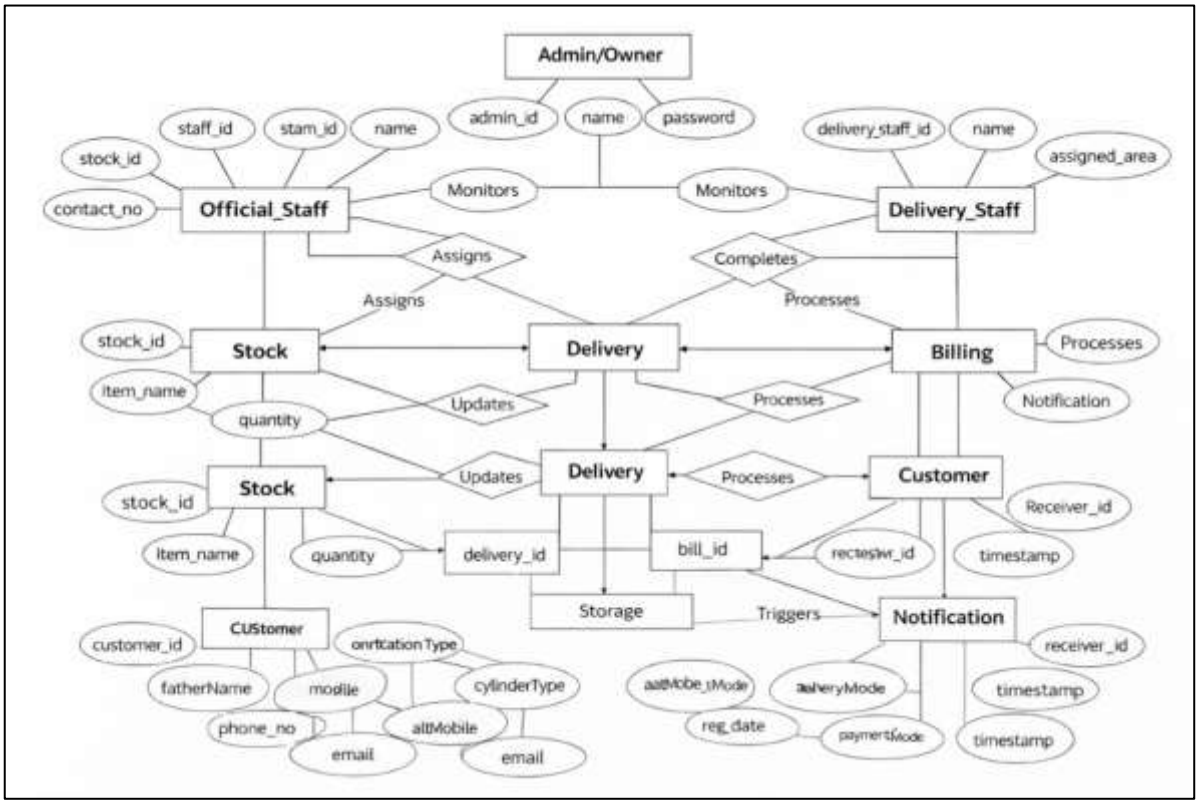


Figure 5.6 E-R diagram of Gas Agency Stock Management system

Figure 5.6, shows the overlook of database where entity shows in rectangle are strong entity, they have its own table where as weak entity is displayed in diamond that are relational table, because they depend on other strong entities for their existence. Oval shows the attribute of every entity. We can say that oval is the field names and rectangle, diamonds show relational table name.

Table 2 Database table description

Sr No.	Table Name	Fields Name	Purpose
1	Admin	Id, name, email, password	Admin table contains agency manager login details.
2	Customer	Id, first name, last name, mobile no, address, connection no, gender	Customer table contains customer personal and connection details.
3	CylinderStock	Id, total cylinder, full cylinder, empty cylinder, update date	Stock table contains cylinder stock details (full/empty/total).
4	Staff	Id, name, role, mobile no, address, joining date.	Staff table contains office and delivery staff details.
5	Delivery	id, customerid, staffid, deliverydate, status, address	Delivery table contains cylinder delivery information.
6	Billing	id, customerid, amount, billdate, paymentstatus, paymentmode	Billing table contains bill and payment details of customer.

7	Booking	id, customer id, booking date, cylinder type, status	Booking table contains gas booking details by customer.
8	Report	id, report type, generate date, details	Report table contains system generated reports (Bills, stock etc.).

Functionality provided by System:

1. **User Management** – Management of admin and staff records.
2. **Stock Management** – Tracking and managing gas cylinder inventory.
3. **Order & Delivery Management** – Handling customer orders and deliveries.
4. **Timeline/Progress Tracking** – Monitoring the status of deliveries and stock movement.
5. **Billing Management** – Generating invoices and tracking payments.

1. User Management:

The User Management module of the Gas Agency Management System handles all the users who interact with the system such as Admin, Staff, and Delivery Person. Each user is assigned specific roles and permissions so that they can access only relevant features of the system.

- Admin has full control over the system (manage stock, billing, staff, delivery).
- Staff can handle billing, customer records, and booking.
- Delivery Person manages cylinder delivery and updates delivery status.

This module helps maintain security, proper data handling, and smooth system operation.

Step wise procedure of user management module:

Step 1. Registration of users through the form

- Admin need to fill details as follows**:** Name, Email, Password.
- Staff need to fill details as follows**:** Name, Mobile Number, Address, Role (Office Staff / Delivery Staff), Joining Date.

- Delivery Person need to fill details as follows**:** Name, Mobile Number, Address, Vehicle Details.

Step 2. Now Admin is able to see all user records (Staff, Delivery Person) on his dashboard.

Step 3. There are many users in the system, so Admin can apply filter options to: Arrange records in ascending and descending order. Search users by name or role. So admin can easily find and view any user data.

Step 4. There are 2 options shown along every user record: Update → Admin can update user details. Delete → Admin can delete user record permanently from the system.

Step 5. Same Step 4 is applicable for: Staff records, Delivery Person records.

Implementation of User Management Module:

Algorithm:

Step 1: Start

Step 2: Create Admin table in database

Step 3: Create Staff table in database

Step 4: Create Delivery table in database

Step 5: Add Admin record through registration form

Step 6: Add Staff record through registration form

Step 7: Add Delivery Person record through registration form

Step 8: Display all Staff and Delivery records in table format on dashboard

Step 9: Add Update and Delete actions for each row in table

Step 10: If Update action is clicked, then:

- Open form
- Get updated data
- Update record in database

Step 11: If Delete action is clicked, then:

- Display confirmation alert
- Delete record if confirmed

Step 12: From Step 8 to Step 11 applicable for:

- Staff records
- Delivery Person records

Step 13: End

2. Stock Management:

The Stock Management module is used to manage the inventory of gas cylinders in the agency. It keeps track of total cylinders, filled cylinders, and empty cylinders available in the stock. Whenever a cylinder is delivered to a customer, the stock is automatically reduced, and when empty cylinders are returned, the stock is updated accordingly. This helps in maintaining accurate stock levels at all times. Admin can also manually update stock in case of any adjustments or errors. The system provides a clear view of stock status on the dashboard, which helps in decision-making. This module ensures that there is no shortage or overstock of cylinders, improving efficiency and reducing operational issues.

Step wise procedure of Stock Management module:

Step 1: Admin enters initial stock details including total cylinders, full cylinders, and empty cylinders.

Step 2: The system stores stock data and displays it on the dashboard for real-time monitoring.

Step 3: When a cylinder is delivered to a customer, the system automatically reduces the count of available/full cylinders.

Step 4: When empty cylinders are returned, the system updates the stock accordingly.

Step 5: Admin can manually update stock values in case of errors or adjustments.

Step 6: The system continuously displays updated stock levels for better decision-making.

Implementation of Stock Management module:

Step 1: Start
Step 2: Create stock table in database
Step 3: Insert initial stock values
Step 4: Display stock details on dashboard
Step 5: On delivery → decrease full cylinders
Step 6: On return → increase empty cylinders
Step 7: Update stock values in database

Step 8: Display updated stock

Step 9: End

3.Order & Delivery Management:

The Order and Delivery Management module handles the complete process of customer booking and gas cylinder delivery. Customers can place orders, which are recorded in the system with a unique booking ID. Admin can view all booking requests and assign delivery staff to each order. The delivery person then delivers the cylinder to the customer and updates the delivery status in the system. The system tracks the entire delivery process from booking to completion. It allows status updates such as Pending, In Progress, Delivered, or Cancelled. This module ensures timely delivery, better coordination between staff, and improved customer satisfaction.

Stepwise procedure of Order and Delivery Management module:

- *From Admin perspective:*

Step 1: Select Order/Booking Management option from dashboard

Step 2: View all customer orders (Pending / Assigned / Delivered)

Step 3: Select specific order for processing

Step 4: Assign delivery staff to selected order

Step 5: After assigning, order details are sent to delivery person

Step 6: Admin can track delivery status

Step 7: If any issue occurs, admin can update or reassign delivery

Step 8: After successful delivery, admin verifies delivery status

Step 9: Admin can view all order history for monitoring

Step 10: End

- *From Delivery Person perspective:*

Step 1: Login and select Delivery option

Step 2: View all assigned delivery orders

Step 3: Select order and check customer details (address, contact)

Step 4: Deliver gas cylinder to customer

Step 5: Update delivery status (Delivered / Not Delivered)

Step 6: If delivery failed, mention reason

Step 7: System updates order status automatically

Step 8: Delivery person can view all completed deliveries

Step 9: End

Implementation of Order and Delivery Management module:

Step 1: Start

Step 2: Add Order Management option in dashboard

Step 3: When user clicks order option, navigate to order management page

Step 4: Fetch all booking/order data from database

Step 5: Display orders in table format with status

Step 6: Provide Assign Delivery Staff option

Step 7: After selecting staff, update staff_id in delivery table

Step 8: Provide delivery dashboard for delivery person

Step 9: Allow delivery person to update delivery status

Step 10: Update status field in database (Pending / Delivered / Failed)

Step 11: Store delivery details and timestamp

Step 12: Display updated status to admin

Step 13: Maintain history of all orders and deliveries

Step 14: End

4. Timeline / Progress Tracking:

The Timeline/Progress Tracking module is used to monitor the overall workflow and activities of the gas agency. It tracks delivery progress, stock updates, and operational performance. This module uses visual indicators such as color codes to represent the status of tasks: Green → Completed, Red → Pending, Blue → In Progress. These indicators help admin quickly understand system status without checking detailed records. It also helps in identifying delays or issues in delivery and stock management, allowing quick corrective actions. This module improves transparency and efficiency in monitoring daily operations.

Stepwise procedure of Timeline / Progress Tracking:

Step 1. The system tracks all operations such as bookings, deliveries, and stock updates.

Step 2. These activities are displayed on the dashboard in an organized format.

Step 3. Color codes are used to represent status:

- Green → Completed
- Red → Pending
- Blue → In Progress

Step 4. Admin monitors the progress of operations daily.

Step 5. Any delay or issue is identified quickly and corrective action is taken.

Implementation of Timeline / Progress Tracking:

Step 1: Start
Step 2: Fetch data from booking, delivery, and stock tables
Step 3: Compare current status with expected status
Step 4: Assign color codes based on condition
Step 5: Display status on dashboard
Step 6: Update dynamically when data changes
Step 7: End

5. Billing Management:

The Billing Management module is responsible for generating bills and managing payment records. After delivery of gas cylinders, a bill is generated for the customer. The system records important details such as customer name, amount, date, and payment method. It also tracks whether the payment is completed or pending. Admin can view all billing records, update them if needed, or delete incorrect entries. This module ensures proper financial tracking and reduces errors in billing. It also helps in maintaining a history of transactions, which is useful for reporting and auditing purposes.

Implementation of Billing Management:

Step 1: After successful delivery, the system generates a bill for the customer.

Step 2: Billing details such as customer name, amount, and date are entered.

Step 3: Admin selects payment mode (Cash / Online).

Step 4: The system stores billing information in the database.

Step 5: Admin can view all billing records in table format.

Step 6: Admin can update or delete billing records if required.

Implementation of Billing Management:

Step 1: Start

Step 2: Create billing table in database with fields: bill_id , customer_id , amount , bill_date payment_status , payment_mode .

Step 3: Add Billing option in admin dashboard menu

Step 4: When admin clicks billing option → Navigate to billing management page

Step 5: Fetch all delivered orders from delivery table Only those orders eligible for billing are displayed

Step 6: Provide Generate Bill button for each delivered order

Step 7: When Generate Bill is clicked → Display billing form , Auto-fill customer details (name, ID) , Enter amount and payment mode

Step 8: Validate input fields Check empty fields and Validate amount format

Step 9: After form submission → Generate unique Bill ID automatically ,Store billing data in billing table

Step 10: Display all billing records in table format Show bill_id, customer_name, amount, date, payment_status

Step 11: Provide Update and Delete options for each record

Step 12: If Update clicked → Open edit form ,Modify billing details Update record in database

Step 13: If Delete clicked → Show confirmation alert ,Delete record if confirmed

Step 14: Manage payment status:If payment completed → mark as "Paid" , If pending → mark as "Unpaid"

Step 15: End

Chapter 6
Result and Application

6.1 Result:

The system developed was tested for our batch of final year diploma in computer engineering. students and the web pages and results are present year.

Figure 6.1 Login Page of Gas Agency Stock Management system

Figure 6.1 : This web page is the landing page of our Gas Agency Stock Management System. It is the main login page for all authorized users. Users need to enter their Username and Password to access the system. Based on their role, users must select the appropriate option such as Owner Login or Staff Login. The background image represents the gas agency environment with gas cylinders and a delivery truck, making the page relevant to the system.

Home page :

Figure 6.2 Home Page of Gas Agency Stock Management system

Figure 6.2 : This web page represents the main dashboard of the Gas Agency Management System. It serves as the central control panel after login, where users can access the major modules of the system. The dashboard contains four important modules: Stock Module, Staff Module, Billing Module, and Delivery Module.

The Stock Module is used to monitor available, filled, and empty cylinders. The Staff Module is designed to maintain delivery staff details and work assignments. The Billing Module allows the generation of customer bills and payment receipts. The Delivery Module helps in managing gas delivery schedules and assignments efficiently.

This dashboard improves system usability by providing a well-organized and user-friendly interface for quick navigation and efficient gas agency operations.

Stock Management Module:

Figure 6.3 Stock Management Module of Gas Agency Stock Management system

Figure 6.3 : This web page shows the Stock Management Module of the Gas Agency Stock Management System, which is used to maintain and monitor gas cylinder stock details. The module provides a dashboard-style overview at the top, displaying key stock statistics such as Total Stock, Full Cylinders, Empty Cylinders, and category-wise stock including Commercial, Domestic, and Industrial cylinders.

The Add Stock button allows users to insert new stock information into the system, while the search bar enables quick searching of stock records based on the company name. The table section below stores stock records in a structured manner with details like Date, Company, Type, and Quantity. This module ensures accurate stock tracking, better inventory control, and efficient management of cylinder availability in the gas agency.

Add Stock From Web Page for Admin :

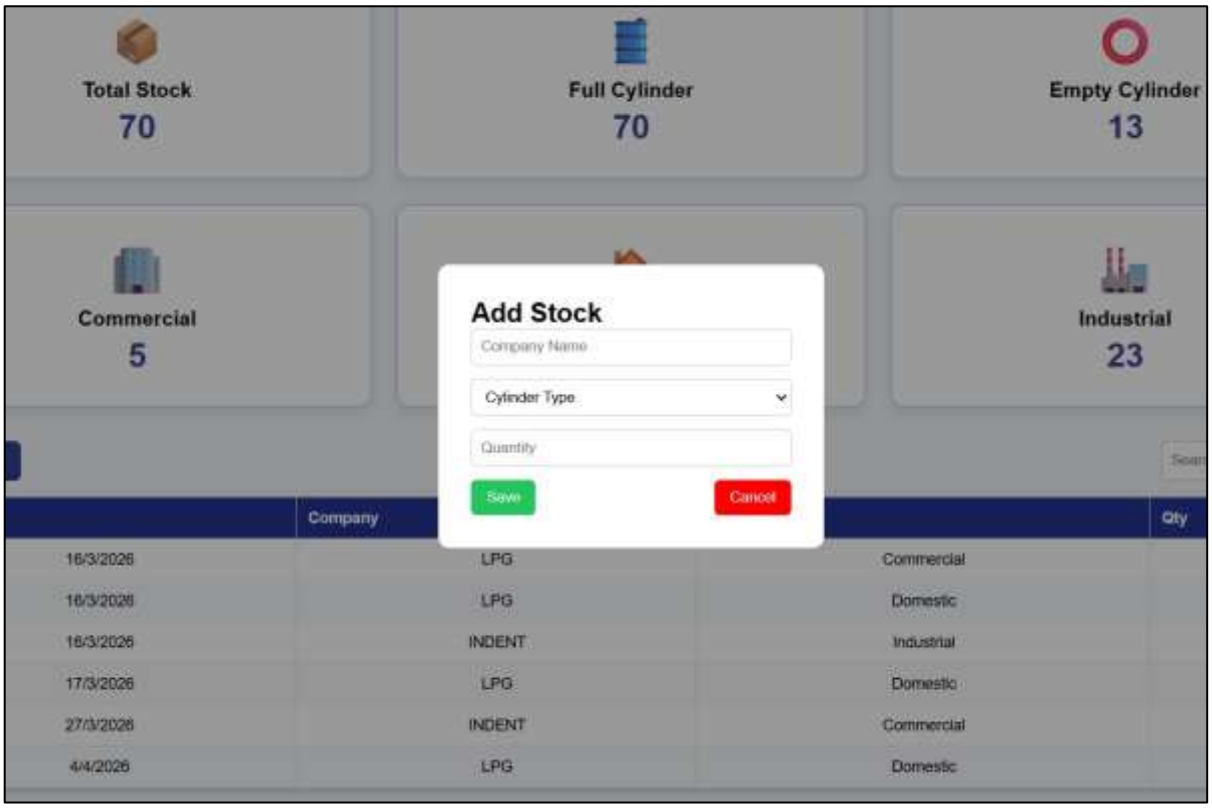


Figure 6.4 Add Stock From for Admin

Stock data is entered through the above Add Stock Form in the system, and the data is filled by the authorized user. This form is used to add new gas cylinder stock details in the Gas Agency Stock Management System. The user needs to enter the Company Name, select the Cylinder Type, and provide the Quantity of cylinders. After entering all details, the user can click on the Save button to store the stock record in the system. The Cancel button is used to close the form without saving the entered data.

Table 3 : Add Stock Form Validations

Fields	Validation Applied
Company Name	"[A-Za-z]+", Only alphabetic characters are allowed
Cylinder Type	Any one cylinder type must be selected such as Commercial, Domestic, or Industrial
Quantity	"[0-9]+", Only numeric values are allowed and quantity must be greater than 0

Delivery Management Module:

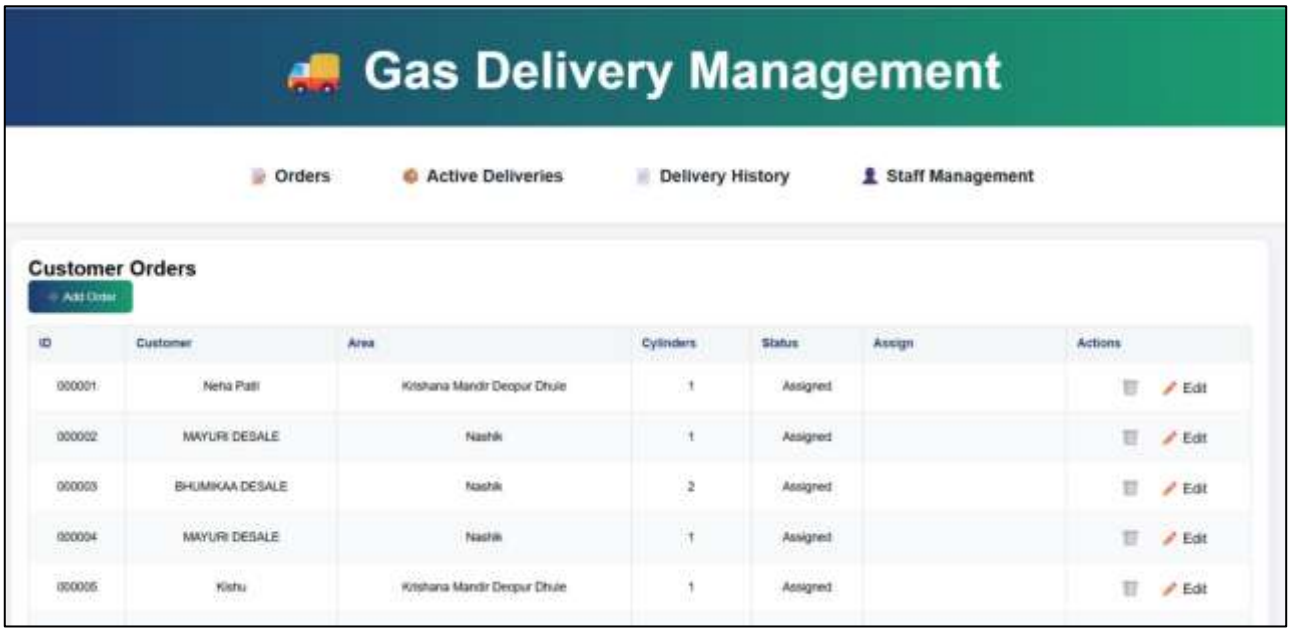


Figure 6.5 Delivery Management Module of Gas Agency Stock Management system

Figure 6.5 : Gas Delivery Management ModuleThis web page shows the Gas Delivery Management Module of the Gas Agency Stock Management System, which is used to manage customer gas cylinder deliveries efficiently. The module provides a navigation dashboard at the top containing options such as Orders, Active Deliveries, Delivery History, and Staff Management for easy access to delivery operations.The Customer Orders section displays all gas cylinder orders placed by customers. The Add Order button allows authorized staff to create new delivery orders in the system. The table below presents order records in a structured format with details such as Order ID, Customer Name, Area, Number of Cylinders, Delivery Status, Staff Assignment, and Actions.Users can assign delivery staff, update delivery status, or edit order information using the available action options. This module helps the gas agency track delivery activities, monitor order status, manage staff assignments, and ensure timely cylinder delivery to customers.

Add Customer Orders From Web Page for Admin :

Figure 6.6 Add Customer Orders From Web Page for Admin

Customer order data is entered through the above **Customer Orders Form** in the system, and the data is filled by the authorized user. This form is used to add customer gas cylinder orders in the **Gas Agency Stock Management System**. The user first enters the **Mobile Number** and clicks on the **Get Customer** button to fetch customer details from the system. After that, the user can verify the **Customer Name**, **Area**, and enter the required number of **Cylinders**. Then the user can click on the **Add Order** button to save the order record in the system. The **Cancel** button is used to clear or close the form without saving the entered data. The table below displays all customer order records with details such as **ID, Customer, Area, Cylinders, Status, Assign, and Actions**.

Table 4 : Customer Orders Form Validations

Fields	Validation Applied
Mobile Number	"[789]\\d{9}", Mobile number must start with 7, 8, or 9 and must be 10 digits
Customer Name	"[A-Za-z]+", Only alphabetic characters are allowed
Area	"[A-Za-z0-9,]+", Area name should contain valid alphabets, numbers, and spaces
Cylinders	"[0-9]+", Only numeric values are allowed and quantity must be greater than 0

Billing Management Module:

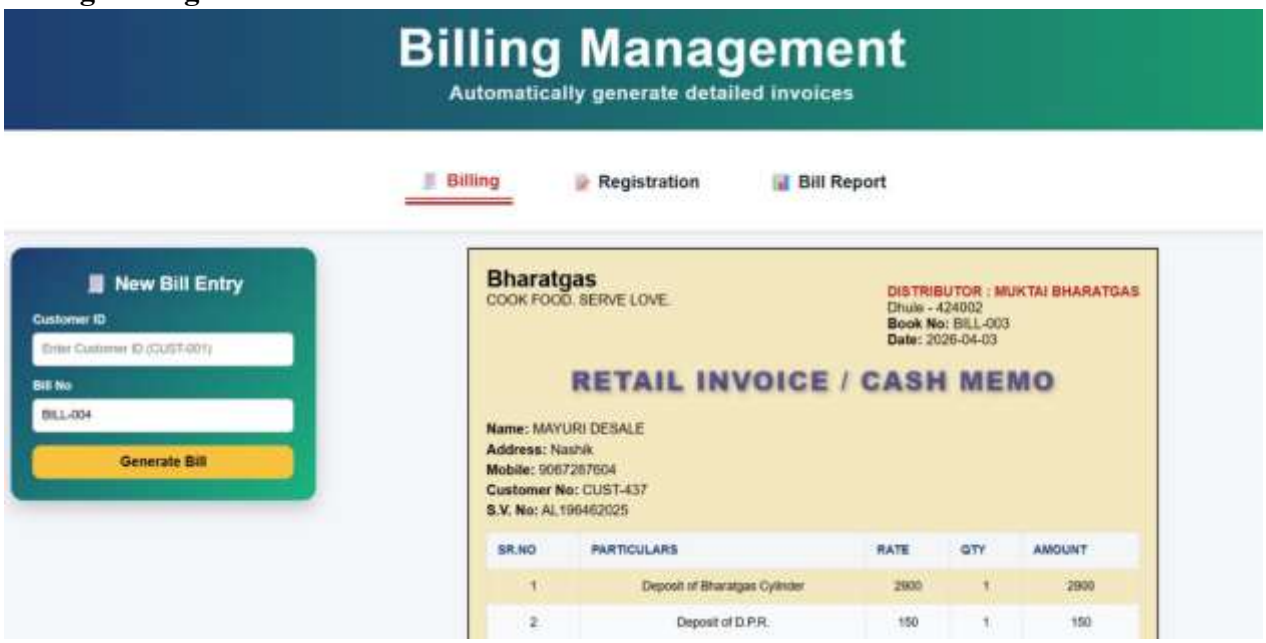


Figure 6.7 Billing Management Module of Gas Agency Stock Management system

Figure 6.7 : Billing Management Module This web page shows the Billing Management Module of the Gas Agency Stock Management System, which is used to generate customer invoices automatically. The module provides navigation options such as Billing, Registration, and Bill Report to manage billing operations efficiently. The New Bill Entry section allows authorized staff to enter the Customer ID and Bill Number to generate a bill. After entering the required details, the user clicks the Generate Bill button, and the system automatically creates a detailed invoice. The invoice section displays complete billing information including Distributor Details, Customer Name, Address, Mobile Number, Customer Number, Bill Date, and Itemized Charges such as cylinder deposit and additional service charges. The system calculates the total amount automatically. This module helps the gas agency maintain accurate billing records, reduce manual calculation errors, and generate professional invoices quickly for customers.

Registration Customer for New Connection From Web Page for Admin :

Figure 6.8 Registration Customer for New Connection From for Admin

The New Gas Connection Registration module is used to register customers applying for a new LPG gas connection in the Gas Agency Stock Management System. Authorized staff enter customer personal details, connection information, cylinder quantity, payment mode, and deposit details through the form. After verification, the data is saved in the system database. This module helps maintain accurate customer records, manage cylinder allocation, and improve agency operations efficiently.

Table 5: New Gas Connection Form Validations

Fields	Validation Applied
Customer ID	Unique ID required (e.g., CUST-001 format)
Customer Name	[A-Za-z]+ – Only alphabetic characters allowed
Father Name	[A-Za-z]+ – Only alphabetic characters allowed
Mobile Number	[789]\\d{9} – Must start with 7, 8, or 9 and contain 10 digits
Alternate Mobile	Optional but must follow mobile number format
Email	Valid email format required (example@gmail.com)
Address	Cannot be empty

City	[A-Za-z]+ – Only alphabets allowed
Pincode	\\d{6} – Must contain 6 digits
Aadhaar Number	\\d{12} – Must contain 12 numeric digits
Connection Type	Selection required
Cylinder Type	Selection required
Cylinders	[0-9]+ – Quantity must be greater than 0
Payment Mode	Selection required
Deposit Amount	Numeric value only and must be greater than 0
Registration Date	Valid date required

6.2 Application of Gas Agency Stock Management System

The Gas Agency Stock Management System is developed to automate and streamline the daily operational activities of a gas agency. The system is used for managing customer registration, gas cylinder stock details, billing processes, delivery tracking, and staff management in an organized and efficient manner. It helps maintain accurate records of cylinder availability, customer orders, and transaction details within a centralized system. By providing real-time updates and automated record maintenance, the system reduces manual paperwork, minimizes operational errors, and improves overall service efficiency. It enables agency staff to monitor stock levels, generate bills quickly, assign deliveries, and track order status effectively. Overall, the system enhances productivity, ensures transparency in operations, and supports smooth management of gas distribution services.

Chapter 7
Testing

7.1 Test Plan:

1. TEST PLAN IDENTIFIER

   Test Plan ID: GAMS-LM-TP-01

2. REFERENCES

   The following documents are referred to for preparing this test plan:

   - Software Requirement Specification (SRS) of Gas Agency Management System
   - Project Synopsis
   - Login Module Design Document
   - Database Design / User Table Structure
   - User Interface Design of Login Screen
   - Validation Rules Document
   - Test Case Document for Login Module

3. INTRODUCTION

   The purpose of this test plan is to verify and validate the **Login Module** of the **Gas Agency Management System**. The login module is one of the most important modules of the system because it controls secure access to the application based on user role.

   This module supports three different login types:

   - **Admin Login**
   - **Official Staff Login**
   - **Delivery Staff Login**

   The test plan ensures that only authorized users can access the system using valid credentials and are redirected to their respective dashboards. It also checks validation of input fields, authentication logic, error handling, and role-based access control.

   The scope of this plan includes functional testing, validation testing, negative testing, and basic role verification of the login process.

4. TEST ITEMS (FUNCTIONS)

The following functions of the **Login Module** will be tested:

- Display of Login Screen
- Username / Email Input Field
- Password Input Field
- Role Selection / Role-based Login Handling
- Login Button Functionality
- Validation for Empty Username Field
- Validation for Empty Password Field
- Validation for Empty Role Selection (if applicable)
- Verification of Correct Credentials for Admin
- Verification of Correct Credentials for Official Staff
- Verification of Correct Credentials for Delivery Staff
- Rejection of Invalid Username or Password
- Rejection of Incorrect Role with Valid Credentials
- Error Message Display for Invalid Login
- Successful Redirection to Admin Dashboard
- Successful Redirection to Official Staff Dashboard
- Successful Redirection to Delivery Staff Dashboard
- Session Handling (basic)
- Logout and Re-login behavior (basic check)

These are the core functional items of the Login Module that will be tested.

5. SOFTWARE RISK ISSUES

The following risk areas are identified in the Login Module:

- Invalid users may gain access if authentication fails
- Wrong role may be assigned after successful login
- Admin may be redirected to the wrong dashboard
- Official Staff or Delivery Staff may access unauthorized features

- Empty fields may be accepted without validation
- Invalid credentials may not show proper error message
- Database connection failure may stop login process
- Password verification may fail due to wrong database mapping
- Application crash may occur during login

Critical Risks

- Authentication failure
- Incorrect role-based access
- Incorrect redirection after login
- Database query failure
- Missing input validation

Since login is the entry point of the system, this module has **High Risk** and must be tested carefully.

6. FEATURES TO BE TESTED

The following user-level features will be tested:

Feature	Risk Level
Login screen opens properly	Medium
Username field accepts valid input	High
Password field accepts valid input	High
Empty field validation	High
Admin login with valid credentials	High
Official Staff login with valid credentials	High
Delivery Staff login with valid credentials	High
Invalid credentials handling	High
Incorrect role access prevention	High
Error message display	Medium
Redirect to correct dashboard after login	High
Basic logout and re-login	Medium

These features are tested from the user's point of view to ensure secure and correct system access.

7. FEATURES NOT TO BE TESTED

The following features are not included in the current Login Module test plan:

- Forgot Password Functionality *(Not implemented in current version)*
- OTP-based Login *(Not included in current release)*
- Biometric / Fingerprint Login *(Future enhancement)*
- Multi-factor Authentication *(Not part of current scope)*
- Advanced Session Timeout Testing *(Outside module scope)*
- Account Lock after multiple failed attempts *(Future enhancement)*

Reason

These features are either not implemented in the current version or planned for future enhancement.

8. APPROACH (STRATEGY)

The testing approach for the Login Module includes:

- **Manual Testing** will be performed on the login screen
- **Functional Testing** will verify login process for Admin, Official Staff, and Delivery Staff
- **Validation Testing** will check empty fields, invalid input, and error messages
- **Negative Testing** will ensure unauthorized users cannot access the system
- **Role-based Testing** will verify correct dashboard redirection for each login type
- **Database Verification** will confirm credentials are matched with correct role records
- **Regression Testing** will be performed after bug fixes

Test Method

The following scenarios will be tested:

- Valid Admin username + valid password
- Valid Official Staff username + valid password
- Valid Delivery Staff username + valid password
- Invalid username + valid password

- Valid username + invalid password
- Valid credentials + wrong role
- Empty username field
- Empty password field
- Both fields empty
- Check error message for invalid login
- Check correct dashboard after successful login
- Check logout and re-login behavior

Tools / Environment

- Android Emulator / Real Mobile Device
- Android Studio
- SQLite / MySQL / Firebase Database (as per project implementation)
- Manual Test Case Sheet

9. ITEM PASS/FAIL CRITERIA

Pass Criteria

The login module will be considered **PASS** if:

- Login screen opens correctly
- Valid Admin credentials allow Admin login
- Valid Official Staff credentials allow Official Staff login
- Valid Delivery Staff credentials allow Delivery Staff login
- Invalid credentials are rejected
- Wrong role access is denied
- Proper error messages are displayed
- Empty fields are not accepted
- User is redirected to the correct dashboard based on role
- System does not crash during login process

Fail Criteria

The login module will be considered **FAIL** if:

- Valid users cannot log in
- Invalid users are allowed access
- Wrong role is allowed access
- Empty fields are accepted
- Wrong or missing error messages are shown
- Redirection fails after successful login
- User is redirected to the wrong dashboard
- Application crashes during login attempt

10. SUSPENSION CRITERIA AND RESUMPTION REQUIREMENTS

Suspension Criteria

Testing of the Login Module will be suspended if:

- Login screen does not load
- Application crashes during login attempt
- Database connection is unavailable
- Authentication query fails completely
- Role-based redirection is not working
- No response is generated after clicking login button

Resumption Requirements

Testing will resume when:

- Login screen is functioning properly
- Database connectivity is restored
- Critical authentication bugs are fixed
- Role-based access issues are resolved
- Updated build/version of the login module is available
- Basic login flow works successfully for re-testing

11. TEST DELIVERABLES

The following deliverables will be produced for Login Module testing:

- Login Module Test Plan

- Login Module Test Cases
- Test Data (valid and invalid credentials for all roles)
- Test Execution Report
- Defect / Bug Report
- Screenshots of Login Results (if required)
- Final Test Summary Report for Login Module

12. REMAINING TEST TASKS

The current test plan covers the core login functionality only. The following related tasks are not fully covered in this phase and may be tested separately in future:

- Forgot Password testing
- OTP verification testing
- Password encryption/security testing
- Account lock after multiple failed attempts
- Session timeout / auto logout testing
- Advanced role permission security testing
- Audit log / login history testing

These tasks are excluded from the current phase because they are either not implemented or beyond the scope of basic login module testing.

13. ENVIRONMENTAL NEEDS

The following environment is required for testing the Login Module:

Hardware Requirements

- Laptop / PC with minimum 4 GB RAM
- Android Mobile Device or Emulator

Software Requirements

- Windows OS / Android OS
- Android Studio
- Database (SQLite / MySQL / Firebase)
- Internet connection (if cloud database is used)

Test Data Requirements

- Valid Admin username and password
- Valid Official Staff username and password
- Valid Delivery Staff username and password
- Invalid usernames
- Invalid passwords
- Empty field inputs
- Wrong role selection inputs (if applicable)

Other Requirements

- User role records must be stored in the database
- Test environment should be stable before execution
- Login screen should be integrated with backend database
- Role-based dashboard screens should be available for validation

14. STAFFING AND TRAINING NEEDS

Staffing Needs

- Student Developer / Tester
- Project Guide / Mentor

Training Needs

- Understanding of login flow and authentication process
- Knowledge of role-based login mechanism
- Understanding of validation rules for username and password
- Basic database verification skills
- Ability to record defects and test results

Since this is an academic project, the student developer will execute the testing under the guidance of the mentor.

15. RESPONSIBILITIES

Student / Developer Responsibilities

- Prepare login module test plan
- Create login test cases for all three roles
- Execute all login test cases

- Verify actual results with expected results
- Report and fix defects
- Re-test corrected issues

Project Guide / Mentor Responsibilities

- Review the login module test plan
- Validate test cases and test results
- Verify role-based access functionality
- Approve successful completion of login module testing

User Responsibilities (if applicable)

- Check usability of login screen
- Verify whether login flow is simple and understandable
- Confirm correct role-based dashboard access

16. SCHEDULE

The testing schedule for the Login Module is as follows:

Testing Activity	Duration	Status
Test Plan Preparation	1 Day	Planned
Test Case Design	1 Day	Planned
Functional Testing for 3 Roles	2 Days	Planned
Validation and Negative Testing	1 Day	Planned
Bug Fixing and Re-testing	1 Day	Planned
Final Test Summary	1 Day	Planned

Schedule Note

Testing will begin after the login module development is completed. If the login screen or database integration is delayed, the test schedule will shift accordingly.

17. PLANNING RISKS AND CONTINGENCIES

Planning Risks

- Login module may not be fully integrated with database
- Delay in completion of login screen development

- Wrong database mapping of username/password/role
- Insufficient test data for all three roles
- Incorrect dashboard linkage
- Last-minute changes in role-based login logic

Contingency Plans

- If database is not ready, UI validation testing will be done first
- If authentication fails, priority will be given to fixing database query logic
- If role-based access fails, redirection logic will be tested separately
- If time is limited, critical test cases (valid login for each role, invalid login, empty field validation) will be executed first
- Additional re-testing will be performed after fixing critical issues

This ensures that the most important login functions are tested even under limited time or resources.

18. APPROVALS

The following persons are responsible for approval of the Login Module testing:

- **Student Developer / Project Owner** – confirms test completion
- **Project Guide / Mentor** – reviews and approves the test results
- **Internal Examiner (if applicable)** – may verify login functionality during project demonstration

Approval Criteria

- All major login test cases are executed
- Admin login works successfully
- Official Staff login works successfully
- Delivery Staff login works successfully
- Invalid login is rejected
- Wrong role access is denied
- Critical defects are fixed
- Login module is ready for system integration

19. GLOSSARY

- **GAMS** – Gas Agency Management System
- **Login Module** – Module used for user authentication and access control
- **Admin** – Main authority user with full system control
- **Official Staff** – Office staff user handling customer records, bookings, and billing
- **Delivery Staff** – Staff user responsible for delivery-related tasks
- **Username** – Unique user identity used for login
- **Password** – Secret credential used for authentication
- **Authentication** – Process of verifying user identity
- **Role-based Access** – Access control based on user type
- **Dashboard** – Main screen shown after successful login
- **Validation** – Checking whether input data is correct and acceptable
- **Test Case** – A condition or scenario used to verify system behavior
- **Defect / Bug** – Error found in the software
- **Pass** – Test executed successfully with expected result
- **Fail** – Test did not produce expected result

7.2 Test Cases:

1. Admin Login Module Test Cases:

TC ID	Test Case Description	Test Data / Input	Expected Result	Actual Result	Status
TC_AL_01	Verify Admin login page opens successfully	Open Admin login page	Admin login page should display properly	As Expected	Pass
TC_AL_02	Verify Admin login with valid credentials	Username: admin01, Password: admin@123	Admin should login successfully and redirect to Admin Dashboard	As Expected	Pass
TC_AL_03	Verify Admin login with	Username: wrongadmin,	System should display invalid username error	As Expected	Pass

	invalid username	Password: admin@123			
TC_AL_04	Verify Admin login with invalid password	Username: admin01, Password: wrong123	System should display invalid password error	**System allowed login with incorrect password**	**Fail**
TC_AL_05	Verify Admin login with both invalid username and password	Username: wrongadmin, Password: wrong123	System should deny login and display error message	As Expected	Pass
TC_AL_06	Verify Admin login when username field is empty	Username: blank, Password: admin@123	System should display "Username is required"	As Expected	Pass
TC_AL_07	Verify Admin login when password field is empty	Username: admin01, Password: blank	System should display "Password is required"	As Expected	Pass
TC_AL_08	Verify Admin login when both fields are empty	Username: blank, Password: blank	System should display validation messages for both fields	As Expected	Pass
TC_AL_09	Verify Admin dashboard redirection after successful login	Valid Admin credentials	Admin Dashboard should open successfully	As Expected	Pass
TC_AL_10	Verify Admin logout functionality	Click logout after login	Admin should logout and return to login page	As Expected	Pass

2. Official Staff Login Module Test Cases:

TC ID	**Test Case Description**	**Test Data / Input**	**Expected Result**	**Actual Result**	**Status**

TC_OSL_01	Verify Official Staff login page opens successfully	Open Official Staff login page	Official Staff login page should display properly	As Expected	Pass
TC_OSL_02	Verify Official Staff login with valid credentials	Username: staff01, Password: staff@123	Official Staff should login successfully and redirect to Staff Dashboard	As Expected	Pass
TC_OSL_03	Verify Official Staff login with invalid username	Username: wrongstaff, Password: staff@123	System should display invalid username error	As Expected	Pass
TC_OSL_04	Verify Official Staff login with invalid password	Username: staff01, Password: wrong123	System should display invalid password error	As Expected	Pass
TC_OSL_05	Verify Official Staff login with both invalid username and password	Username: wrongstaff, Password: wrong123	System should deny login and display error message	As Expected	Pass
TC_OSL_06	Verify Official Staff login when username field is empty	Username: blank, Password: staff@123	System should display "Username is required"	As Expected	Pass
TC_OSL_07	Verify Official Staff login when password field is empty	Username: staff01, Password: blank	System should display "Password is required"	**No validation message displayed for empty password field**	**Fail**
TC_OSL_08	Verify Official Staff login when both fields are empty	Username: blank, Password: blank	System should display validation messages for both fields	As Expected	Pass
TC_OSL_09	Verify Staff dashboard	Valid Staff credentials	Official Staff Dashboard	As Expected	Pass

	redirection after successful login		should open successfully		
TC_OSL_10	Verify Official Staff logout functionality	Click logout after login	Official Staff should logout and return to login page	As Expected	Pass

3. Delivery Staff Login Module Test Cases

TC ID	**Test Case Description**	**Test Data / Input**	**Expected Result**	**Actual Result**	**Status**
TC_DSL_01	Verify Delivery Staff login page opens successfully	Open Delivery Staff login page	Delivery Staff login page should display properly	As Expected	Pass
TC_DSL_02	Verify Delivery Staff login with valid credentials	Username: delivery01, Password: delivery@123	Delivery Staff should login successfully and redirect to Delivery Dashboard	As Expected	Pass
TC_DSL_03	Verify Delivery Staff login with invalid username	Username: wrongdelivery, Password: delivery@123	System should display invalid username error	As Expected	Pass
TC_DSL_04	Verify Delivery Staff login with invalid password	Username: delivery01, Password: wrong123	System should display invalid password error	As Expected	Pass
TC_DSL_05	Verify Delivery Staff login with both invalid username and password	Username: wrongdelivery, Password: wrong123	System should deny login and display error message	As Expected	Pass
TC_DSL_06	Verify Delivery Staff login when username field is empty	Username: blank, Password: delivery@123	System should display “Username is required”	As Expected	Pass

TC_DSL_07	Verify Delivery Staff login when password field is empty	Username: delivery01, Password: blank	System should display "Password is required"	As Expected	Pass
TC_DSL_08	Verify Delivery Staff login when both fields are empty	Username: blank, Password: blank	System should display validation messages for both fields	As Expected	Pass
TC_DSL_09	Verify Delivery dashboard redirection after successful login	Valid Delivery Staff credentials	Delivery Dashboard should open successfully	**System redirected to wrong page after login**	**Fail**
TC_DSL_10	Verify Delivery Staff logout functionality	Click logout after login	Delivery Staff should logout and return to login page	As Expected	Pass

7.3 Defect Report:

1. Admin Login Module:

Field	**Description**
Defect ID	DEF-001
Title / Summary	Admin Login Accepts Invalid Password
Description	When the Admin user enters a valid username with an incorrect password, the system still allows login and redirects the user to the Admin Dashboard. This is a critical authentication issue because invalid credentials should never be accepted. This defect may lead to unauthorized system access and compromise security.
Steps to Reproduce	1. Launch the Gas Agency Management System application. 2. Navigate to the Admin Login page. 3. Enter a valid Admin username (e.g., **admin01**). 4. Enter an incorrect password (e.g., **wrong123**). 5. Click on the **Login** button.

	6. Observe the system response.
Expected Result	The system should reject the login attempt and display an **invalid password error message**.
Actual Result	The system accepts the invalid password and redirects the user to the **Admin Dashboard**.
Environment Details	**Device:** Desktop / Laptop**OS:** Windows 10 / Windows 11 **Browser/App:** Web Application / Android App **Database:** MySQL **Technology:** Java / PHP / Firebase / Android Studio
Severity	Critical
Priority	High
Module/Feature Affected	Admin Login Module / Authentication
Reported By	Tester
Reported On	April 08, 2026
Attachments	Login screen screenshot and Admin Dashboard screenshot after invalid password login
Status	New
Assigned To	Developer
Comments/ Notes	This defect is highly critical because it violates authentication rules and may result in unauthorized admin access. The password validation logic should be reviewed and corrected immediately before deployment.

2. Official Staff Login Module:

Field	**Description**
Defect ID	DEF-002
Title / Summary	Password Validation Message Not Displayed in Official Staff Login
Description	When the Official Staff user leaves the password field blank and attempts to log in, the system does not display the required validation message. The login process does not provide proper feedback to the user,

	which may cause confusion and reduce usability. This defect affects form validation behavior in the Official Staff Login Module.
Steps to Reproduce	1. Launch the Gas Agency Management System application. 2. Navigate to the Official Staff Login page. 3. Enter a valid username (e.g., staff01). 4. Leave the password field blank. 5. Click on the Login button. 6. Observe the validation response.
Expected Result	The system should display a validation message such as **"Password is required"** and should not allow login.
Actual Result	No validation message is displayed for the empty password field, and the user is not properly informed about the missing input.
Environment Details	**Device:** Desktop / Laptop **OS:** Windows 10 / Windows 11 **Browser/App:** Web Application / Android Ap p**Database:** MySQL **Technology:** Java / PHP / Firebase / Android Studio
Severity	Medium
Priority	Medium
Module/Feature Affected	Official Staff Login Module / Form Validation
Reported By	Tester
Reported On	April 08, 2026
Attachments	Login screen screenshot showing empty password field and no validation message
Status	New
Assigned To	Developer
Comments/Notes	This defect affects usability and input validation standards. Although it may not block the system completely, it should be corrected to improve user guidance and maintain proper validation consistency across all login modules.

3. Delivery Staff Login Module:

Field	Description
Defect ID	DEF-003
Title / Summary	Delivery Staff Login Redirects to Incorrect Dashboard
Description	When the Delivery Staff user enters valid login credentials and clicks on the login button, the system authenticates successfully but redirects the user to an incorrect dashboard instead of the Delivery Staff Dashboard. This issue affects role-based access and may lead to unauthorized screen visibility or confusion for the user.
Steps to Reproduce	1. Launch the Gas Agency Management System application. 2. Navigate to the Delivery Staff Login page. 3. Enter valid Delivery Staff credentials (e.g., delivery01, delivery@123). 4. Click on the Login button. 5. Observe the redirected screen after successful authentication.
Expected Result	After entering valid Delivery Staff credentials, the system should redirect the user to the **Delivery Staff Dashboard**.
Actual Result	The system authenticates the Delivery Staff user successfully but redirects the user to the **Official Staff/Admin Dashboard** instead of the Delivery Staff Dashboard.
Environment Details	**Device:** Desktop / Android Device **OS:** Windows 10 / Android 13 **Browser/App:** Web Application / Android App **Database:** MySQL **Technology:** Java / PHP / Firebase / Android Studio
Severity	High
Priority	High
Module / Feature Affected	Delivery Staff Login Module / Role-Based Authentication
Reported By	Tester

Reported On	April 08, 2026
Attachments	Screenshots of login screen and incorrect dashboard redirection
Status	New
Assigned To	Developer
Comments / Notes	This defect indicates improper role-based redirection after successful authentication. The login validation is working correctly, but the post-login navigation logic is assigning the wrong dashboard to the Delivery Staff role. This issue should be fixed before deployment because it impacts user access flow and system usability.

Chapter 8
Conclusion and Future Scope

7.1 Conclusion:

In summary, the Gas Agency Management System developed in this project serves as an efficient and user-friendly platform for managing all major operations of a gas agency. The system is designed to simplify and automate tasks for Admin and Staff, ensuring smooth handling of daily activities. The Admin can effectively manage user records, monitor stock levels, handle customer orders, assign delivery staff, and track overall progress. The system also provides a structured approach to managing billing operations, ensuring accurate invoice generation and payment tracking. Staff members can efficiently perform their assigned roles such as handling bookings, managing deliveries, and updating system records. The delivery process is streamlined, allowing real-time status updates and better coordination between different users. With features like role-based access, real-time tracking, and organized data management, the system enhances operational efficiency and reduces manual effort. The use of simple interfaces and clear workflows makes the system easy to use and understand. Overall, this system improves productivity, ensures better resource management, and provides a reliable solution for gas agency operations. By focusing on accuracy, efficiency, and ease of use, the system helps in delivering better service and achieving organizational goals effectively.

7.2 Future Scope:

In future, the Gas Agency Management System can be enhanced by adding features like real-time delivery tracking, online booking and digital payment options for customers. Notification services such as SMS or email alerts can also be integrated to provide updates on booking and delivery status. Additionally, an automatic stock alert system can be implemented to notify low inventory levels. A mobile application version and advanced reporting with data analytics can further improve efficiency, decision-making, and overall user experience.

Reference and Bibliography

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- **Websites**

1. W3Schools – HTML, CSS, PHP, Bootstrap Tutorials
2. MDN Web Docs – Web Development Concepts
3. GeeksforGeeks – DBMS & System Design
4. TutorialsPoint – PHP & MySQL Tutorials
5. StarUML – ER Diagram & UML Design

2. Bibliography:

The development of the Gas Agency Stock Management System Website is based on the study of existing manual and semi-digital gas agency processes. The system focuses on improving stock management, billing generation, delivery tracking, and customer record handling through a web-based solution.